BASED ON

Spike Neural Network

For Recognition on Neural Signals

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FreeRider no 1
FreeRider no 2

Table of Contents

Sections :

Background

1851	1865+	1984	Modern	Future
Spring and Metal Tendons	Inserting hinges at the knees and ankles	The first microprocessor prosthesis	Limitation of muscle actions in swinging	**High-performance feedback and control system**

Based on IntechOpen (2024). Trade-Plus Technologies: A Review. Source accessed April 2026.

Problems

Prosthetic body part is missing something!

The Computational Barrier in smart prosthetics is that Traditional Neural Networks are too computationally expensive to run on portable, battery-powered prosthetic devices, creating a trade-off between intelligence and practicality

Spike Neural Network (SNN) is a solution to this problem!

Johnny Silverhand - Cyberpunk 2077

Winter Soldier - Marvel

Nero - Devil May Cry 5

Traditional Neural Network (CNN)

LIMITS

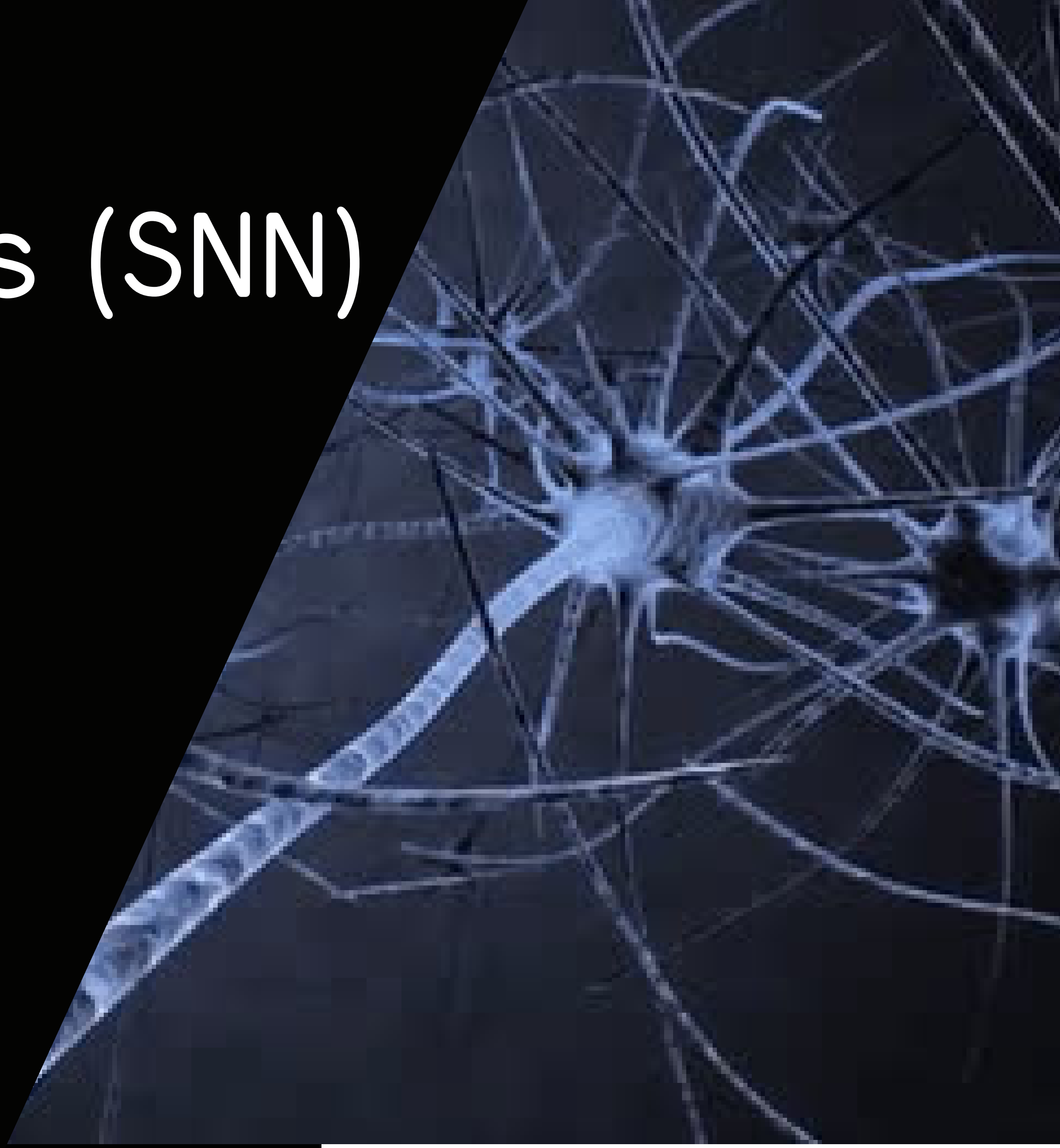
- High Energy
- **Dense floating-point** operations per layer
- **Requires** CPU/GPU **resources**
- NOT Wearable

Prosthetic Needs

REQUIRE

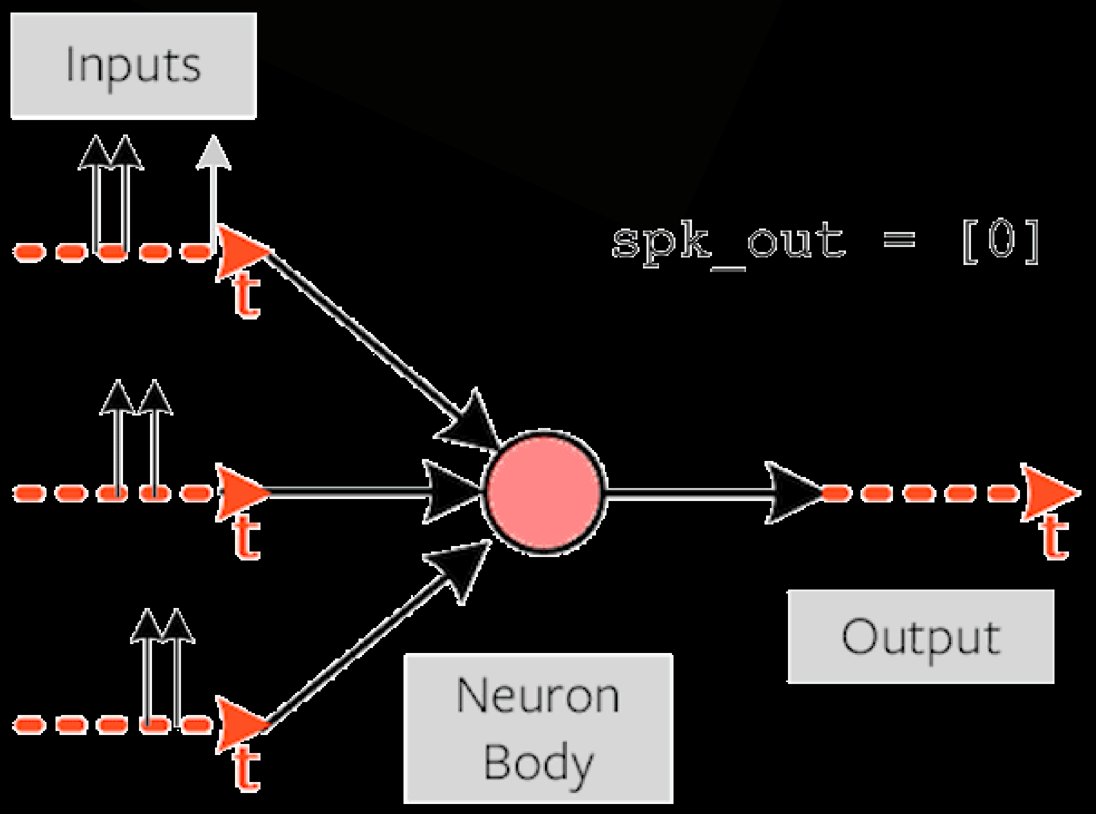
- Ultra-low **Power**
- Real-time **Response**
- **Lightweight hardware**
- **Always ON**, portable

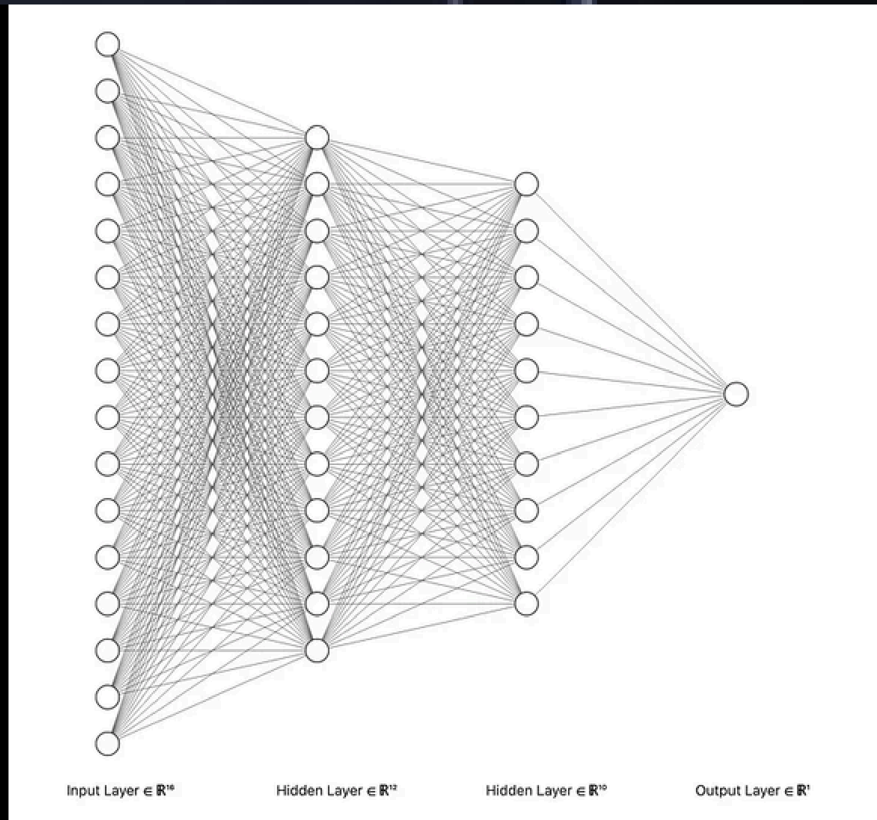
Our Solution: Spiking Neural Networks (SNN)



Biologically Inspired, Energy Efficient

- SNNs process data as sparse spike events — only compute when a spike fires
- Inspired by how the biological brain works
- Key advantage: event-driven computation → massive power savings
- Uses Leaky Integrate-and-Fire (LIF) neurons

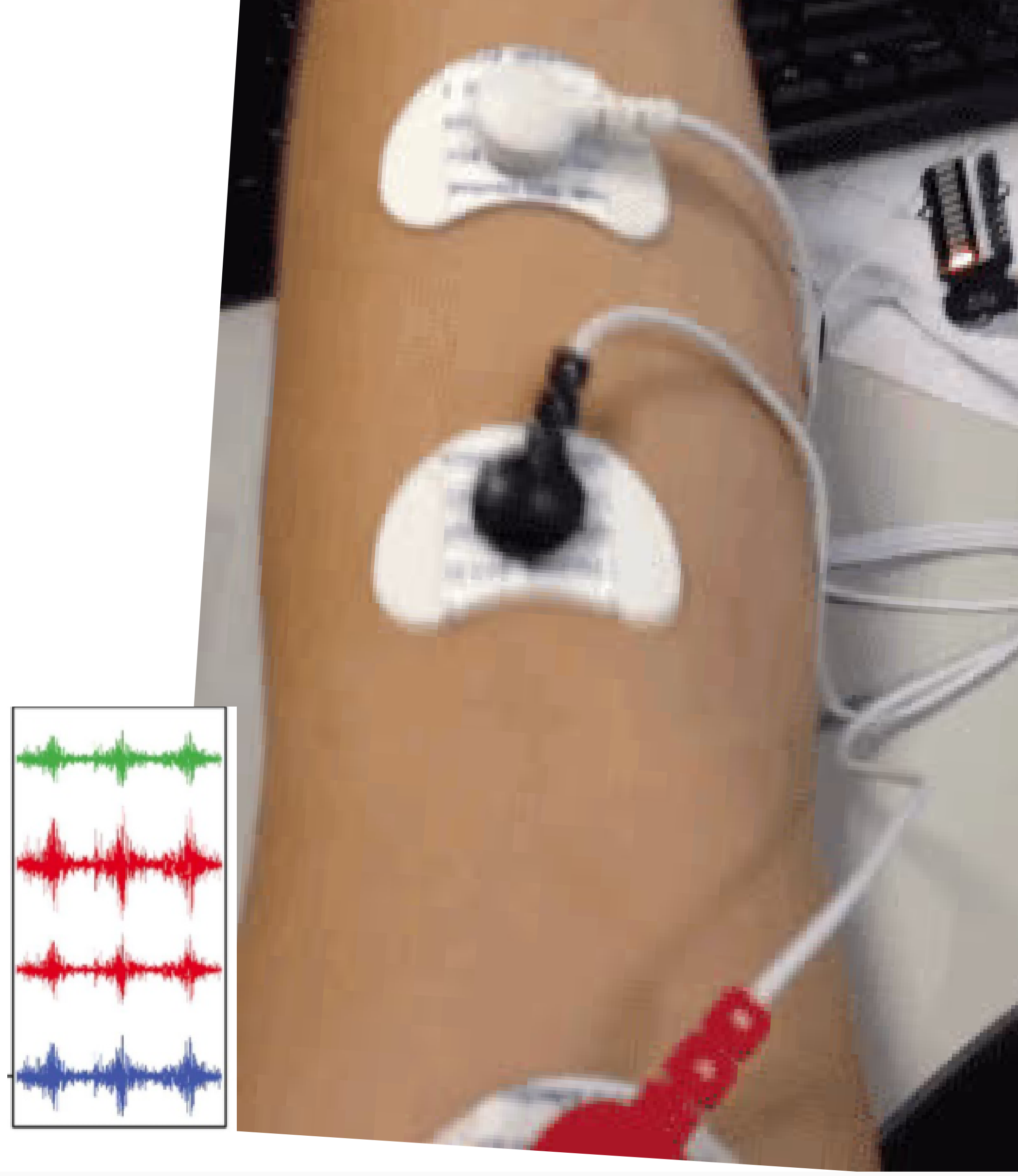


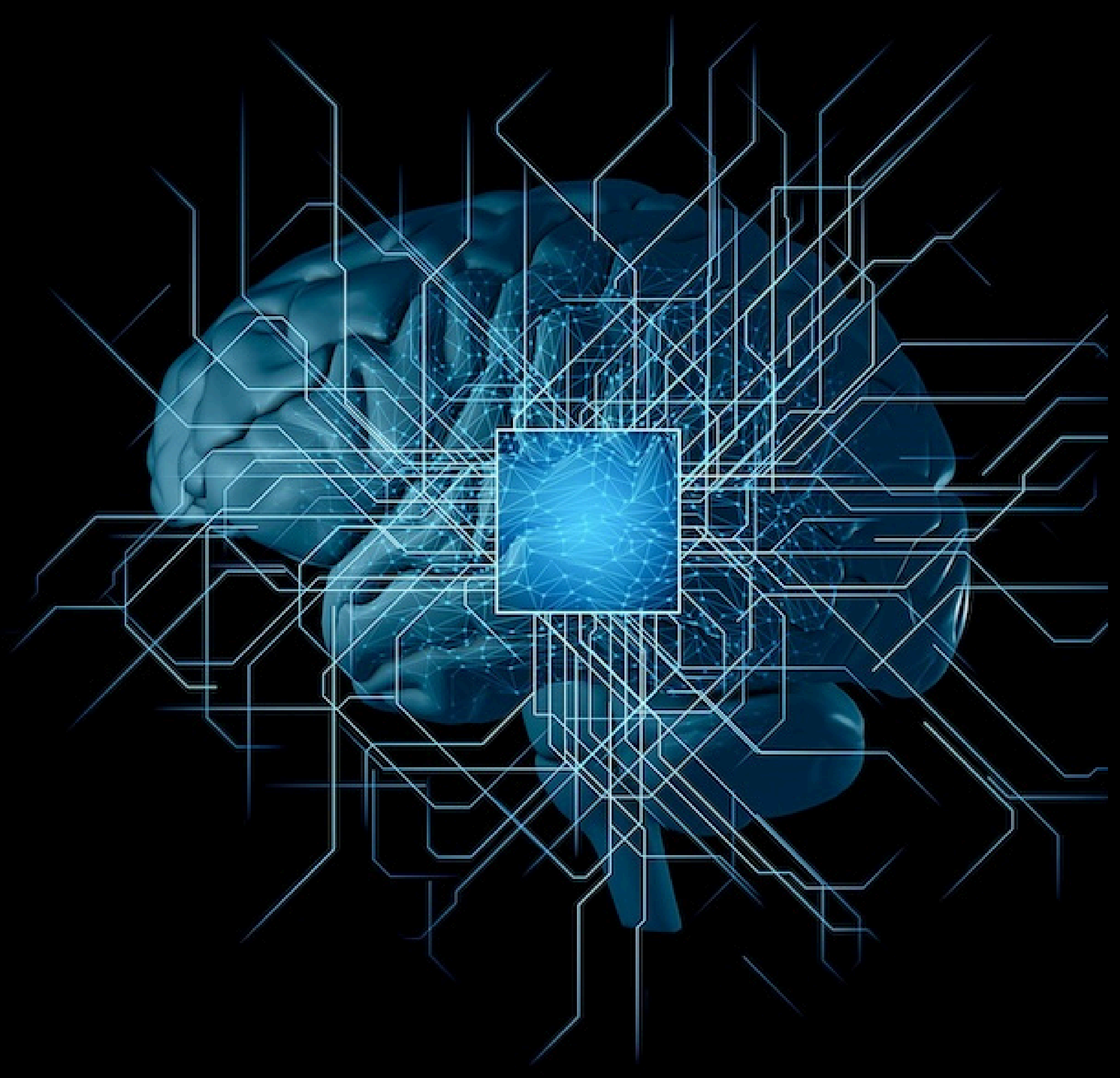


Dataset & sEMG Signals

NinaPro-Data Base, a standardized database for hand movement recognition

Using NinaPro DB1 for training and testing. It has 27 subject with 10 channels of sEMG signals. We train our model on exercise 2 that contains 17 hand gestures + a rest state.





Model Development

SNN model

Input Stage (Translator)

Takes messy muscle signals from the NinaPro sensors and processes them into a digital pattern that the spiking network can interpret.

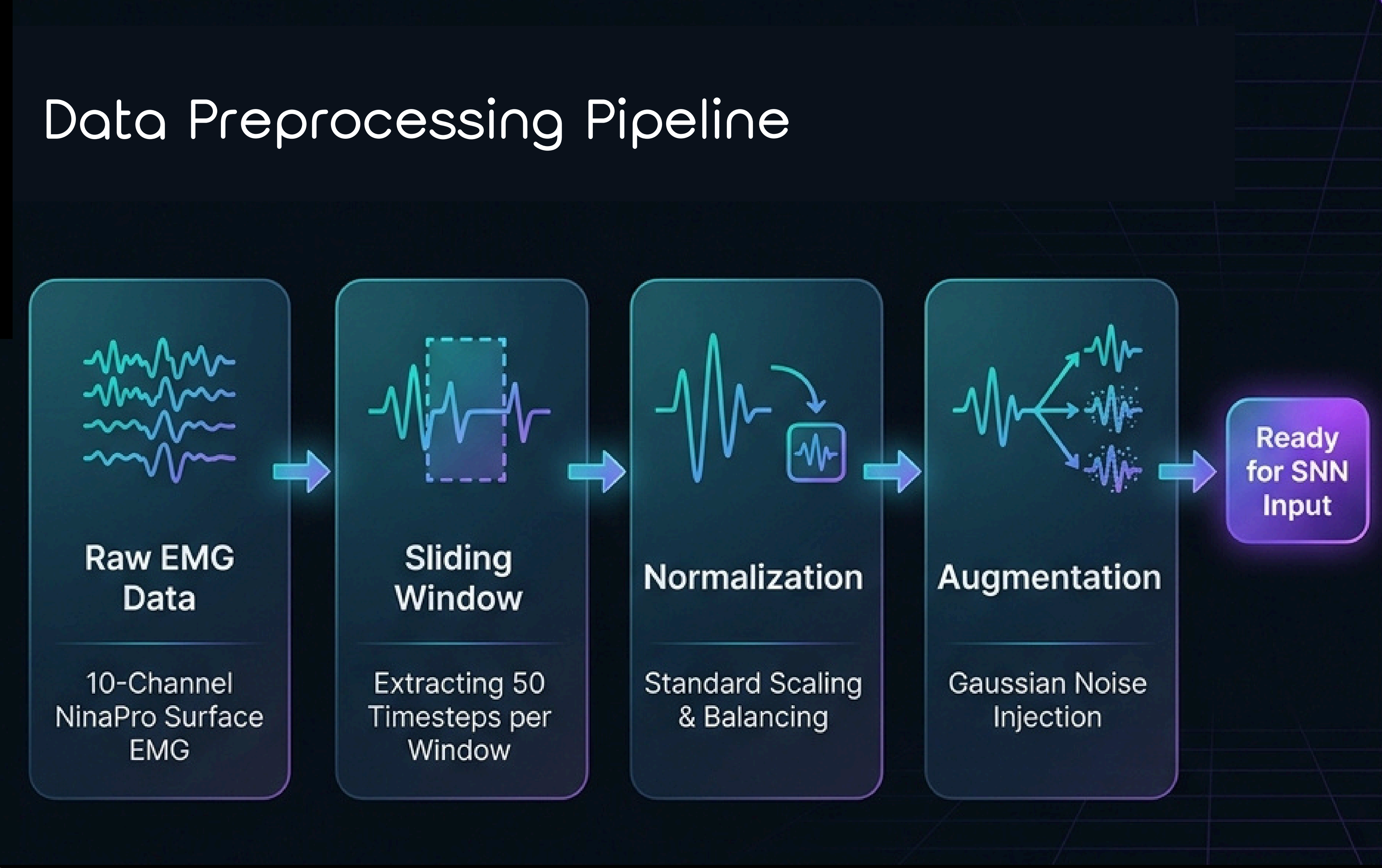
The Hidden Layers (3-Layer)

Uses spiking neurons that only fire when necessary, allowing it to recognize complex movements while minimizing power consumption.

Output Stage (Decision Maker)

Analyzes the network's spiking patterns and triggers a hand movement only when a high-confidence intent is detected.

Dataset Preprocess



01 **Windowing**

Raw sequences are segmented into fixed-length windows of 50 timesteps

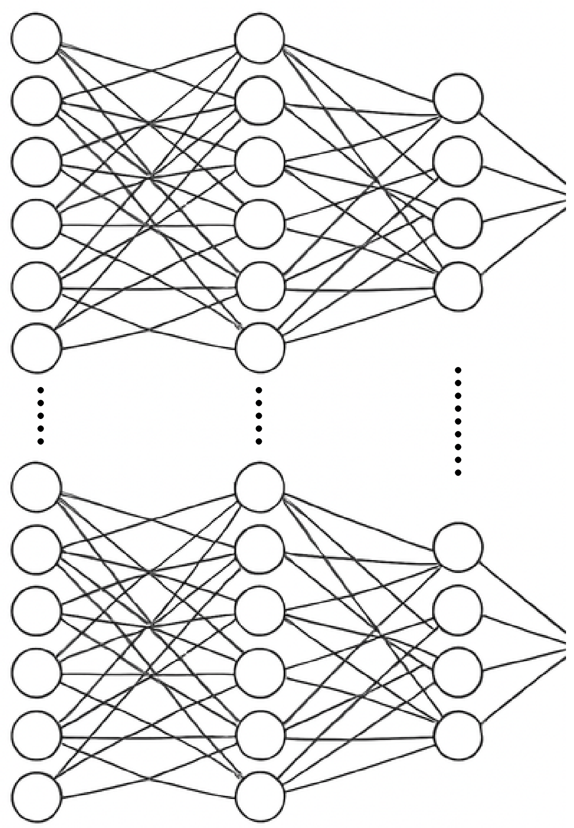
02 **Normalization**

Applied Standard Scaling to zero-center the data and achieve unit variance.

03 **Augmentation**

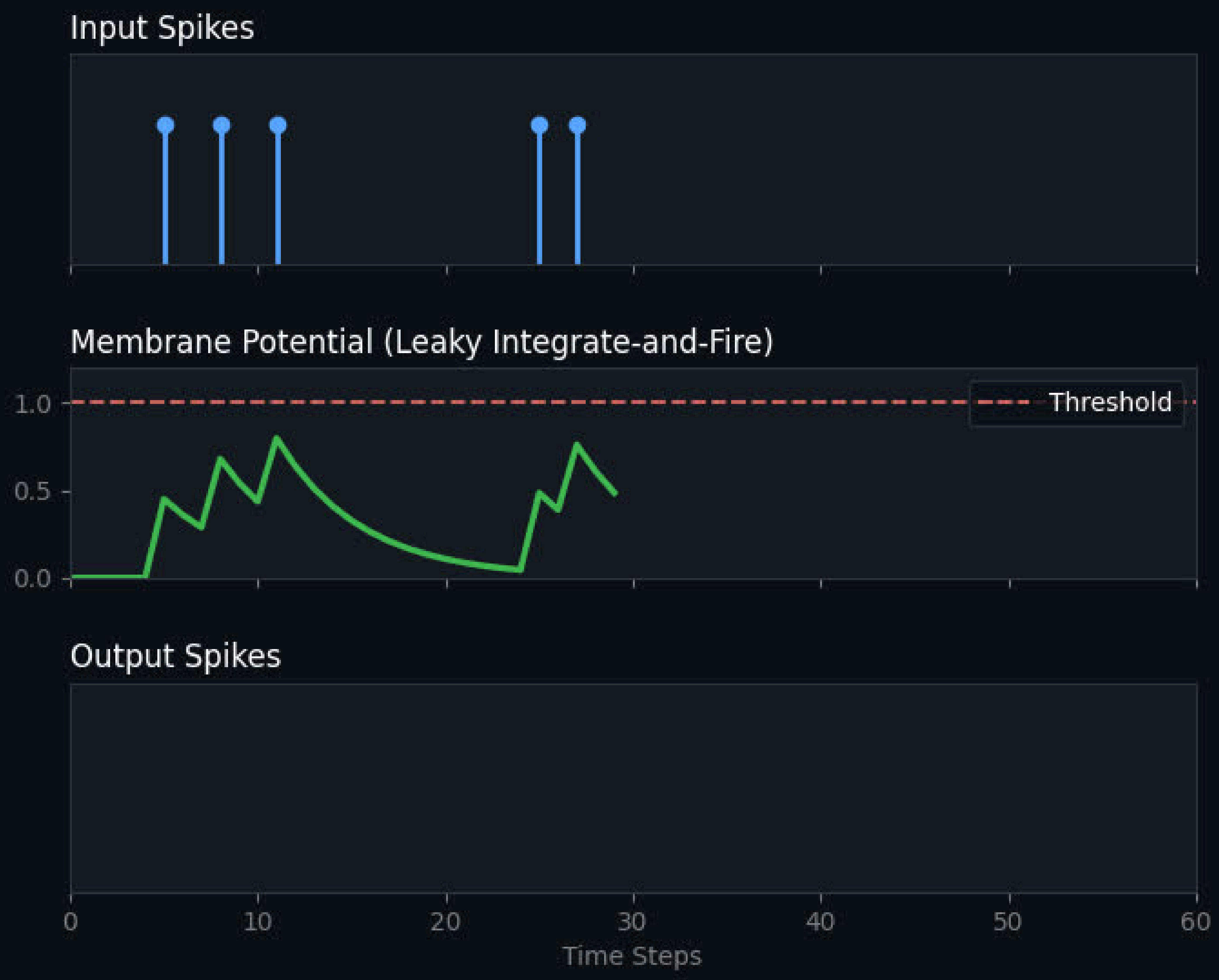
Injected stochastic Gaussian Noise into the training set to enhance model robustness.

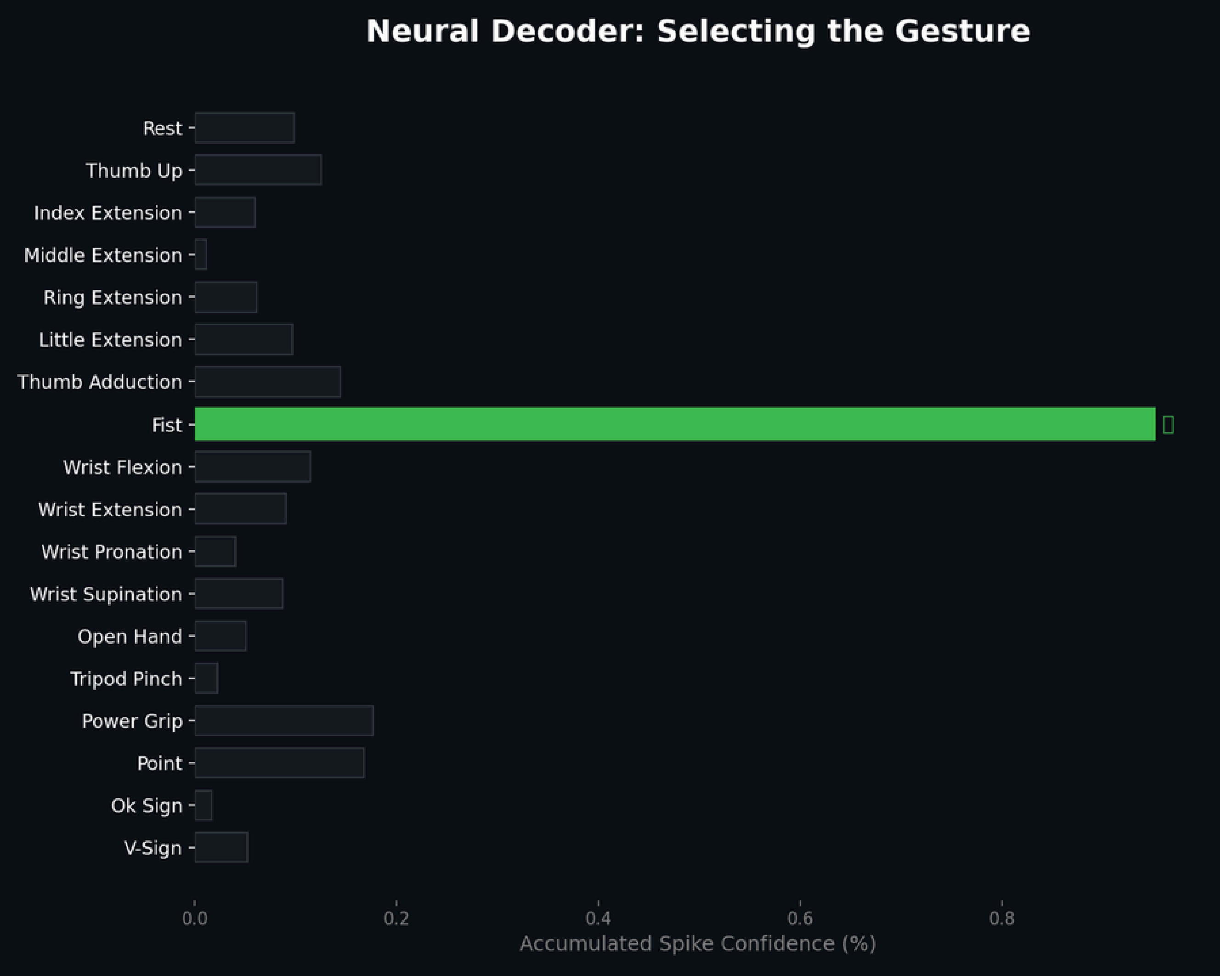
The Hidden Layers

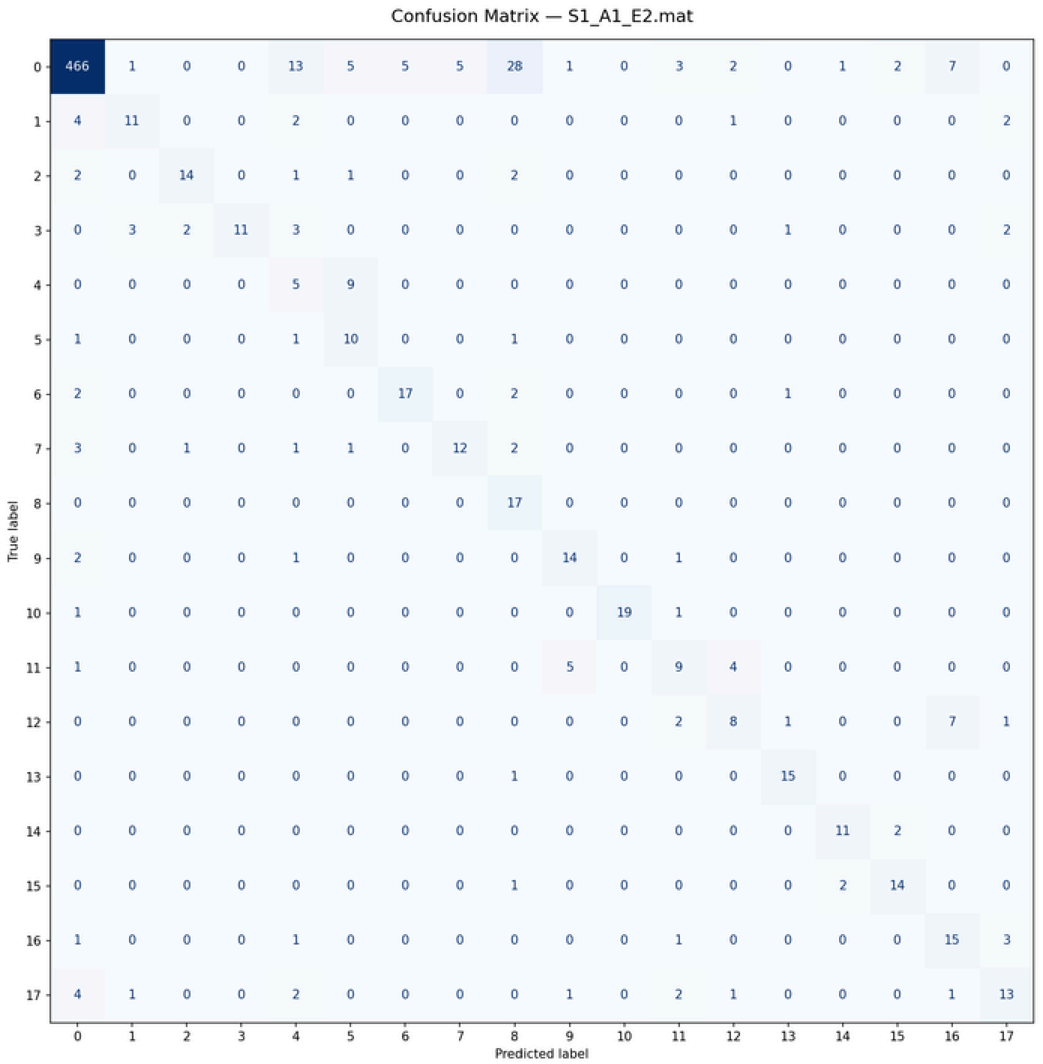


256 - 256 - 128
(LIF Neurons per layer)

Leaky Integrate-and-Fire neurons







Output layer

- Cumulative Integration

  Each of the 18 gesture classes is assigned a dedicated output neuron.

  The system integrates (sums) spikes across the 50-timestep window to measure gesture probability.

- Winner-Takes-All Decision

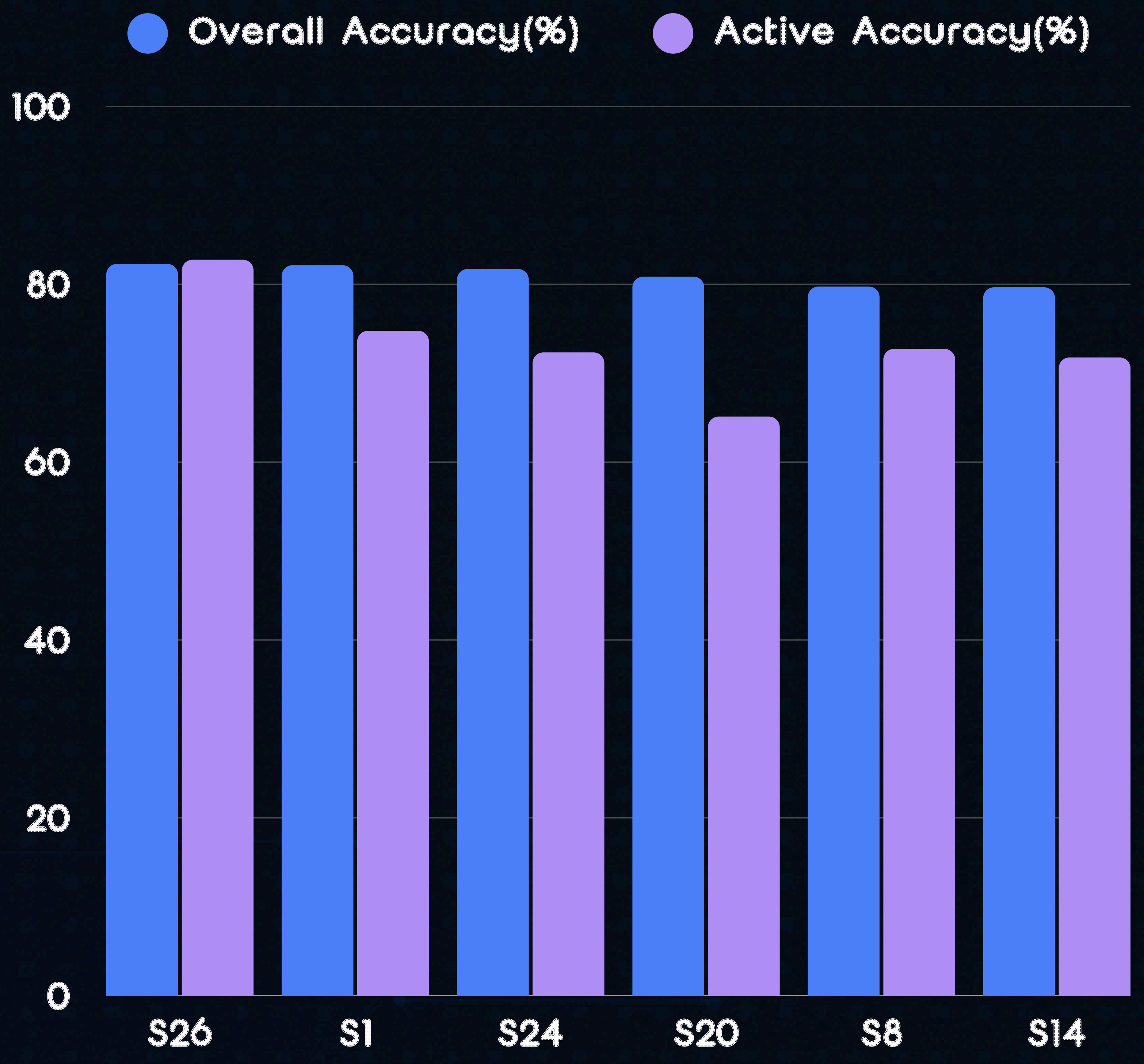
  We apply an Argmax function—the neuron with the highest accumulated spike count is selected as the intended movement.

- High-Speed Execution

  Achieves a final decision in as little as 1.95ms (on optimal subjects), for exceeding the 300ms threshold for perceived real-time human reaction.

Result & Compare

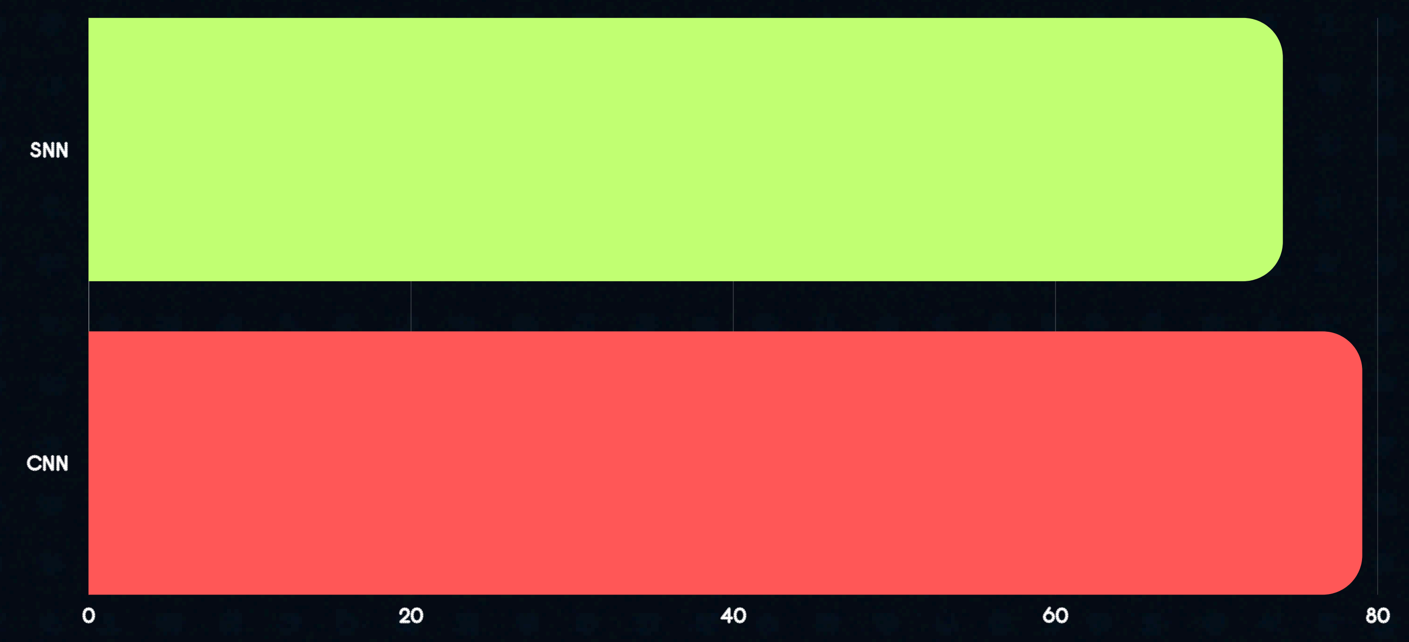
Classification Accuracy Across 27 Subjects



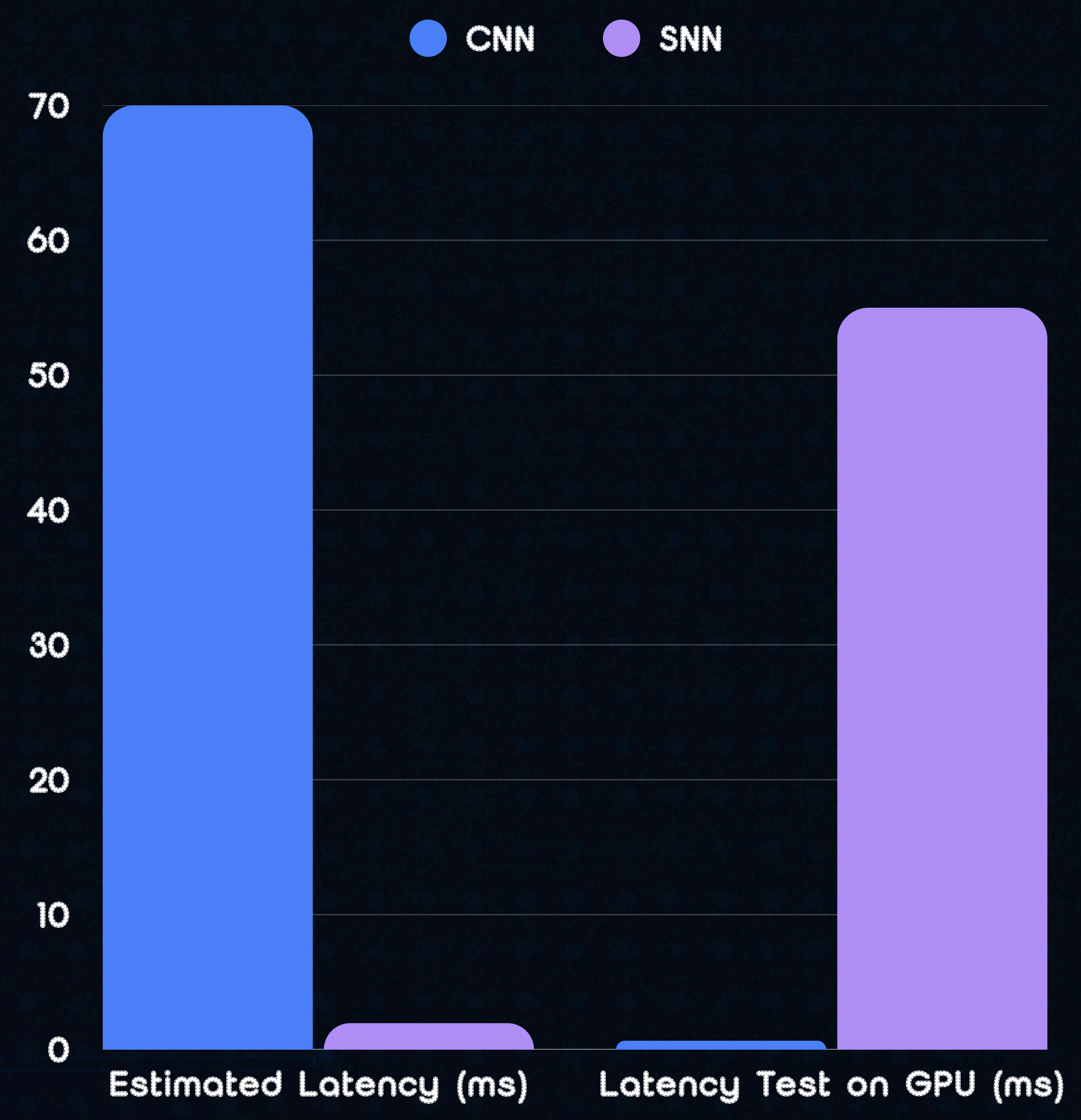
Accuracy Result on SNN

- Average Overall Accuracy: 74.98%
- Average Active Gesture Accuracy: 69.46%
- Best Performance: S26 (overall accuracy:82.28%, active accuracy:82.75%)

Compare to 1D-CNN (Overall Accuracy)



Latency Result



Estimated Latency represents

- execution on event-driven hardware (e.g., Intel Loihi) where only spikes trigger compute.
- And the CNN running on a low-power microcontroller or mobile CPU (non-GPU).

which estimates a 36.3x energy reduction.

Comparsion (Hardware Estimates)

Metric	SNN (Estimated Neuromorphic)	CNN (Estimated Edge GPU)
Latency	~1-5ms (e.g. Intel Loihi)	~20-50ms
Power Draw	1-10 mW (Milliwatts)	5-15+ W(watts)
Compute Paradigm	Event-driven (Sparse, spikes)	Always-on (Dense float operations)
Protability	Highly wearable, small battery	Tethred, requires bulky battery
Accuracy	~75.6%	~82.8%

SNN consumes 100 TIMES less power than CNN !

Current Limitations

Simulated Environment

Simulation on GPUs lead to restricted performance metrics compared to real neuromorphic hardwares.

Biological Uniqueness

Low generalized performance and high signal variability due to subject-specific physiological traits, i.e. body fat percentage, skin conductance.

Future works based on limitations

Cross-subject Transfer Learning

Implement pre-trained model to many users. And adjusting the weights to individual user physiology

Neuromorphic Deployment

Deploying the SNN model onto specialized event-driven processors / neuromorphic chip

Hardware Integration

Integrating with real prosthetic hardware prototype, allowing real-time validation and continuous feedback loops

Resource

GiLberT-Mak/**FYP**

1 Contributor · 0 Issues · 0 Stars · 0 Forks

GiLberT-Mak/FYP

Contribute to GiLberT-Mak/FYP development by creating an account on GitHub.

GitHub

Image/GiF source : From internet and genrated by AI

Q&A !!